PART 1 - GENERAL

1.1 Summary

26 33 53 Part 1.1.1.A This section specifies the requirements for modular static uninterruptible power supply (UPS) systems, battery energy storage cabinets, and associated accessories for the Meridian Data Centre Campus, Phase 1. The UPS systems shall provide continuous, conditioned power to the IT server load across four independent Server Halls.

26 33 53 Part 1.1.1.B The scope of supply includes UPS power modules, static bypass switches, battery cabinets, monitoring and control interfaces, and all interconnecting power and control cabling within the UPS room.

1.2 References

26 33 53 Part 1.2.1.A The following standards shall apply: IEC 62040-1 (General and Safety Requirements), IEC 62040-2 (Electromagnetic Compatibility), IEC 62040-3 (Performance Requirements and Test Methods), IS 16242 (UPS Requirements), IEC 60896 (Stationary Lead-Acid Batteries).

1.3 Submittal Requirements

26 33 53 Part 1.3.1 The Contractor shall submit the equipment submittal package referencing the current active project specification revision. The active specification revision for this section is Rev 2024. Submittals referencing an incorrect or outdated specification revision shall be returned without review.

26 33 53 Part 1.3.2.A Submittals shall include certified manufacturer product data sheets, performance test reports, dimensional drawings with weights, single line diagrams, battery sizing calculations, and a clause-by-clause compliance matrix.

1.4 Quality Assurance

26 33 53 Part 1.4.1.A UPS manufacturer shall have a minimum of 15 years experience in the design and manufacture of static UPS systems rated above 500 kW, with at least 5 reference installations in Tier III or above data centres.

26 33 53 Part 1.4.1.B Manufacturing facility shall hold current ISO 9001 and ISO 14001 certifications.

26 33 53 Part 1.4.1.C UPS equipment shall be tested and certified in accordance with IEC 62040-3 (Method of Specifying the Performance and Test Requirements). The Contractor shall submit IEC 62040-3 type test reports from an accredited testing laboratory as part of the submittal package.

1.5 Warranty and Spares

26 33 53 Part 1.5.1.A The UPS manufacturer shall provide a minimum of 24 months on-site warranty from the date of commissioning. Warranty shall include all power electronic components, control boards, fans, and capacitors.

26 33 53 Part 1.5.1.B A recommended spare parts list with pricing shall be submitted for Consultant review. Minimum spares to be supplied include one complete set of control boards, one fan assembly, and one set of DC fuses per UPS group.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers and System Configuration

26 33 53 Part 2.1.1.A Acceptable UPS manufacturers: VoltEdge Power Systems, or approved equivalent meeting all performance requirements of this section.

26 33 53 Part 2.1.2.A UPS system modules shall be configured in a minimum N+2 parallel redundant configuration to ensure system availability during concurrent maintenance and single fault conditions. Each Server Hall shall be served by an independent UPS group.

2.2 Performance and Design Criteria

26 33 53 Part 2.2.1.A Each UPS module shall have a minimum continuous output power rating of 1200 kW at unity power factor (1.0 PF), 415 VAC three-phase output, 50 Hz. Rating shall be verified at 40°C ambient temperature.

26 33 53 Part 2.2.1.B UPS module output power factor shall be unity (1.0 PF) at full rated load. No derating for power factor below unity is permitted.

26 33 53 Part 2.2.1.C UPS operating topology shall be double-conversion online mode (VFI-SS-111) per IEC 62040-3 classification. The UPS shall provide continuous voltage and frequency regulation independent of utility supply conditions.

26 33 53 Part 2.2.1.D Input total harmonic current distortion (THDi) shall not exceed 5.0% at full rated load without external input harmonic filters. THDi measurement shall be per IEC 62040-3 Annex E.

26 33 53 Part 2.2.2.A Double-conversion online mode efficiency at 100% rated load shall be a minimum of 96.0%. Efficiency shall be measured per IEC 62040-3 at nominal input voltage and rated load.

26 33 53 Part 2.2.2.B Double-conversion online mode efficiency at 75% and 50% rated load shall each be a minimum of 96.0%. All efficiency values shall be measured in VFI mode with harmonic filters active.

26 33 53 Part 2.2.3.A Output voltage regulation shall be within +/- 1% of nominal under steady state conditions, and within +/- 5% during 100% load step transients, with recovery to +/- 1% within 20 milliseconds.

26 33 53 Part 2.2.4.A Output frequency regulation shall be within +/- 0.1% of 50 Hz under free-running conditions.

26 33 53 Part 2.2.5.A UPS module shall sustain 125% of rated load for a minimum of 10 minutes continuously without transfer to static bypass. At 150% overload, the UPS shall sustain the load for a minimum of 30 seconds before bypass transfer.

26 33 53 Part 2.2.6.A Input power factor shall be 0.99 or better at full rated load.

26 33 53 Part 2.2.7.A ECO mode efficiency shall be a minimum of 99.0% at rated load. ECO mode transfer to double-conversion mode shall occur within 2 milliseconds of detecting input power disturbance.

26 33 53 Part 2.2.8 The static bypass switch transfer time shall not exceed 4 milliseconds for any transfer between inverter output and bypass supply, measured as total interruption duration per IEC 62040-3 classification.

2.3 Battery and Energy Storage System

26 33 53 Part 2.3.1.A Battery system shall use Valve Regulated Lead-Acid (VRLA) batteries in sealed maintenance-free construction mounted in dedicated battery cabinets.

26 33 53 Part 2.3.1.B Battery system shall comprise 240 cells of 2 V nominal (40 blocks of 12 V batteries in series) providing a nominal DC bus voltage of 480 VDC.

26 33 53 Part 2.3.1.C Battery string configuration shall consist of a minimum of 6 parallel strings of 200 Ah (C10 rate) battery blocks per UPS module.

26 33 53 Part 2.3.1.D Battery monitoring system shall include individual block voltage monitoring, string current monitoring, ambient and pilot cell temperature sensors, and Modbus TCP/IP interface for integration with the BMS.

26 33 53 Part 2.3.1.E The battery system shall support the full rated UPS load at 100% capacity for a minimum continuous runtime of 10.0 minutes. Battery sizing calculations shall account for the high-rate discharge factor and end-of-life capacity degradation of 20%.

26 33 53 Part 2.3.2.A Total battery bank nominal energy capacity shall be a minimum of 576 kWh at the nominal DC bus voltage of 480 VDC, equivalent to a minimum of 1200 Ah at the C10 rate. Vendor datasheet shall clearly state both kWh and Ah capacity values.

26 33 53 Part 2.3.2.B Battery runtime performance shall be rated and verified at a load power factor of 1.0 PF (unity). Runtime claims at reduced power factors shall not be accepted as equivalent.

26 33 53 Part 2.3.2.C Battery system nominal DC bus voltage shall be 480 VDC to match the UPS inverter DC link requirements. End-of-discharge voltage shall not fall below 408 VDC (1.7 V per cell).

26 33 53 Part 2.3.3 Battery shelf life without charge shall be a minimum of 6 months from the date of manufacture without permanent capacity loss. The vendor shall submit storage conditions and shelf life test data as part of the submittal.

26 33 53 Part 2.3.4 UPS double-conversion efficiency shall be a minimum of 96.0% at each of the 50%, 75%, and 100% load points, measured in VFI mode per IEC 62040-3. Efficiency values shall be based on active power (kW) measurements.

26 33 53 Part 2.3.5.A Maximum floor-load weight per fully loaded battery cabinet shall not exceed 2800 kg. Floor loading calculations based on cabinet footprint and weight distribution shall be submitted for structural review.

26 33 53 Part 2.3.5.B Battery cabinet dimensions shall permit standard freight elevator transport. Maximum individual module shipping weight shall be noted on the dimensional drawing.

26 33 53 Part 2.3.6.A Battery design life shall be a minimum of 10 years at 25°C average ambient temperature per IEC 60896-21 / Eurobat classification.

26 33 53 Part 2.3.6.B Battery system shall maintain full manufacturer warranty coverage and published performance parameters across an ambient operating temperature range of 0°C to 25°C.

26 33 53 Part 2.3.7.A Battery cabinets shall include hydrogen gas detection sensors with alarm contacts connected to the BMS. Ventilation calculations for the battery room shall be submitted per IEC 62485-2.

26 33 53 Part 2.3.8 Battery cabinet power cable entry shall accommodate both top and bottom cable entry conduits to match overhead cable tray or under-floor cable routing as determined by the final installation layout.

26 33 53 Part 2.3.9.A Each battery cabinet shall incorporate a manual maintenance bypass switch allowing isolation of individual battery strings without disrupting the remaining parallel strings.

26 33 53 Part 2.3.9.B Battery cabinet DC distribution shall include appropriately rated fuses or circuit breakers on each battery string for short circuit protection.

26 33 53 Part 2.3.9.C A Low Voltage Disconnect (LVD) contactor shall be provided on each battery cabinet to prevent deep discharge damage. The LVD contactor ampere rating and breaking capacity shall be submitted with the battery cabinet specification sheet.

2.4 Physical Construction and Enclosure

26 33 53 Part 2.4.1.A UPS module enclosure shall be constructed of minimum 1.6 mm steel sheet with IP20 minimum ingress protection rating per IEC 60529.

26 33 53 Part 2.4.1.B Enclosure shall include lockable front and rear access doors with captive fasteners. Minimum service clearance of 1000 mm shall be maintained on all accessible sides.

26 33 53 Part 2.4.2 UPS module cooling airflow shall be configured for front-inlet and top-exhaust to support hot aisle/cold aisle containment architecture in the UPS room. Rear-exhaust or side-exhaust configurations are not acceptable.

26 33 53 Part 2.4.3.A UPS module cooling fans shall be hot-swappable without requiring UPS shutdown or load transfer. Fan failure alarms shall be reported via the monitoring interface.

26 33 53 Part 2.4.4.A All internal power connections shall utilise bolted joints with calibrated torque values marked on the assembly. Spring-type terminal connections are not permitted for power circuits above 100 A.

26 33 53 Part 2.4.5.A UPS module enclosure shall have a seismic rating suitable for Zone III per IS 1893 or equivalent IBC classification.

26 33 53 Part 2.4.5.B Equipment shall be suitable for installation on a raised access floor with standard pedestal height of 600 mm. Anti-vibration pads shall be provided under all mounting points.

26 33 53 Part 2.4.5.C UPS module enclosure paint finish shall be RAL 7035 (Light Grey) polyester powder coating, minimum 60 micron average dry film thickness, with corrosion resistance suitable for indoor C1 environment per ISO 12944.

PART 3 - EXECUTION

3.1 Examination and Preparation

26 33 53 Part 3.1.1.A Prior to delivery, the Contractor shall verify that the UPS room structural slab, cable trays, and HVAC provisions are complete and ready to receive equipment.

3.2 Installation

26 33 53 Part 3.2.1 Verify all power cable terminations have been torque-marked per manufacturer specifications. Inspect structural mounting bolts and anti-vibration pads for Module 1. Record torque values on the commissioning checklist.

26 33 53 Part 3.2.2 Verify all power cable terminations have been torque-marked per manufacturer specifications. Inspect structural mounting bolts and anti-vibration pads for Module 2. Record torque values on the commissioning checklist.

26 33 53 Part 3.2.3 Verify all power cable terminations have been torque-marked per manufacturer specifications. Inspect structural mounting bolts and anti-vibration pads for Module 3. Record torque values on the commissioning checklist.

26 33 53 Part 3.2.4 Verify all power cable terminations have been torque-marked per manufacturer specifications. Inspect structural mounting bolts and anti-vibration pads for Module 4. Record torque values on the commissioning checklist.

26 33 53 Part 3.2.5 Verify all power cable terminations have been torque-marked per manufacturer specifications. Inspect structural mounting bolts and anti-vibration pads for Module 5. Record torque values on the commissioning checklist.

3.3 Field Quality Control

26 33 53 Part 3.3.1.A Perform insulation resistance testing on all power cables prior to energisation. Minimum acceptable insulation resistance is 100 MΩ at 1000 VDC for LV circuits.

3.4 Commissioning and Acceptance Testing

26 33 53 Part 3.4.1.A The commissioning agent shall perform a complete functional test of each UPS module including input/output voltage verification, frequency stability, and battery charger operation.

26 33 53 Part 3.4.2.A Verify UPS double-conversion efficiency is a minimum of 96.0% at both 100% and 75% rated load points. Efficiency measurements shall use calibrated power analysers with accuracy class 0.5 or better.

26 33 53 Part 3.4.2.B Perform input harmonic current measurement at full rated load. Input THDi shall not exceed the value specified in Part 2.2.1.D of this section.

26 33 53 Part 3.4.2.C During site battery discharge testing, the inverter shall maintain full 100% IT load for a minimum of 12 minutes on battery power. The discharge test shall be conducted at design ambient temperature with all battery strings connected.

26 33 53 Part 3.4.2.D Perform UPS static bypass transfer test. Verify seamless transfer from inverter to bypass and back with no load interruption exceeding the specified transfer time.

26 33 53 Part 3.4.2.E Verify UPS module parallel operation and active load sharing. Load imbalance between parallel modules shall not exceed 5% of rated load.

26 33 53 Part 3.4.2.F Simulate single module failure and verify N+1 redundancy takeover. Remaining modules shall assume full load without interruption.

26 33 53 Part 3.4.2.G Perform 100% load step test and verify output voltage transient recovery is within the limits specified in Part 2.2.3.A of this section.

26 33 53 Part 3.4.2.H Verify battery recharge time from fully discharged state. Battery system shall reach 90% of nominal capacity within 8 hours of recharge initiation.

26 33 53 Part 3.4.2.I Perform BMS communication integration test. Verify all UPS alarm and status signals are correctly mapped and displayed on the central BMS operator console.

26 33 53 Part 3.4.2.J Execute a coordinated full-load transfer test from utility to generator and back. Verify the UPS maintains uninterrupted output power throughout the transfer sequence.